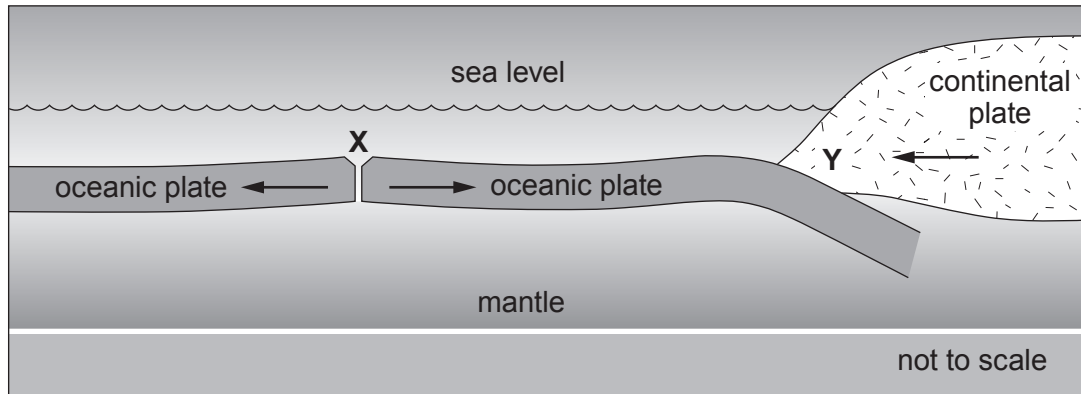


4. The Earth's crust is broken up into tectonic plates.

The diagram shows two plate boundaries **X** and **Y**.



(a) Describe what happens as the plates move apart at plate boundary **X**.

[2]

.....

.....

.....



- (b) The two types of plate have different densities as shown in the table.

Type of plate	Density (g/cm ³)
continental	4.7
oceanic	5.0

At boundary **Y** the oceanic plate is pushed **underneath** the continental plate.

- (i) State why the oceanic plate is pushed underneath the continental plate. [1]

.....

- (ii) State what happens to the oceanic plate when it goes underneath the continental plate. [1]

.....

- (iii) Underline the name of the type of plate boundary seen at **Y**. [1]

destructive

constructive

conservative

- (c) Using satellite imaging, scientists discovered that one oceanic plate had moved 537 cm over 600 years. Calculate the mean rate of movement of the plate over this time. [2]

$$\text{rate} = \frac{\text{distance}}{\text{time}}$$

Rate = cm/year

